WEEK 1-2

QUESTION 1.

Consider the following strategic-form simultaneous game. Mork's payoffs are listed first.

		Mindy		
		X	Y	Z
Mork	A	80 , 30	0 , 20	20 , 20
	B	85 , 15	20 , 20	5 , 5
	C	95 , 25	5 , 25	5 , 25

a. Does either player have a dominant strategy? If yes, list it/them.

b. Does either player have a dominated strategy? If yes, list it/them.

c. List all pure strategy Nash equilibria of the above game?

d. If Mindy's payoffs in the top-left box (A,X) increased from 30 to 300, how would your answer to part c above change? Explain.

QUESTION 2.

You and another firm are engaged in a joint venture. Each year, you and the other firm decide whether to invest effort in the joint venture (which is costly), or simply free-ride on the effort of the other firm. The game is represented below.

		Other Firm	
		Effort	No Effort
You	Effort	25 , 25	0 , 50
	No Effort	50 , 0	10 , 10

a. Formally confirm that this game is a prisoner's dilemma.

b. Imagine that you expect to play this game with the Other Firm each year and forever. The Other Firm employs a grim trigger strategy, cooperating each year only if you have been cooperative in every previous year. What must be true of the interest rate (numerically, what must the range of interest rates be) for cooperation to be sustainable?

c. In one or two sentences, given your answer in part c., remark on how likely it is that cooperation will be sustained.

d. Imagine now that, if you play non-cooperatively, the Other Firm will not be able to observe this for two years. That is, the other firm still plays the grim trigger strategy, but does not begin punishment for non-cooperative behavior until two years later. Does this make cooperation more or less likely? Briefly explain.

QUESTION 3.

Two friends are deciding where to go for dinner. There are three choices, which we label A, B, and C. Max prefers A to B to C. Sally prefers B to A to C.
To decide which restaurant to go to, the friends adopt the following procedure:

First, Max eliminates one of three choices.

Then, Sally decides among the two remaining choices.

Thus, Max has three strategies (eliminate A, eliminate B, and eliminate C). For each of those strategies, Sally has two choices (choose among the two remaining).

a. Write down the extensive form (game tree) to represent this game.

b. If Max acts non-strategically, and makes a decision in the first period to eliminate his least desirable choice, what will the final decision be?

c. What is the equilibrium of the above game?

d. Does your answer in b. differ from your answer in c.? Explain why or why not.

QUESTION 4.

Consider the following game

		Player 2		
		Left	Middle	Right
Player 1	Up	**1**, 2	**3**, 5	**2**, 1
	Middle	**0**, 4	**2**, 1	**3**, 0
	Down	**-1**, 1	**4**, 3	**0**, 2

a) Does either player have a dominant strategy? Explain.

b) Does either player have a dominated strategy? Explain.

c) Solve the equilibrium for this game.

QUESTION 5.

Two mountain hotels in Carpathians – Zakhar Berkut and Veza Vedmeza, compete for tourists from three different cities in Ukraine: Kyiv, Lviv, and Odesa. The capacity of both hotels is sufficient for the accommodation of all tourists in only one of them. Both hotels have financial resources for the advertising campaign in only one city, the effectiveness of their campaigns is the same. If only one hotel runs the campaign in a given city, it gains all tourists from this city and hence the following profit: in the case of Kyiv 150 thousand HRN, in the case of Lviv 90 thousand HRN and in the case of Odesa 72 thousand HRN. If both hotels run the campaign in the same city, they receive the half of the profit from this city each. Similarly, in the case that no hotel runs the campaign in a specified city.

a. What are the optimal strategies for both hotels?

b. Suppose, Veza Vedmeza brand is more popular and in the case of the campaign or absence of campaign in the same city 70% of profit goes to VV and 30% to ZB. What are the optimal strategies for both hotels?

QUESTION 6.

An investor wants to build two hotels. One of them will be called Big (abbreviated as B); gaining the order for it will bring the profit of 15 million to the building firm. The second hotel will be called Small (abbreviated as S); gaining the order for it will bring the profit of 9 million to the building firm. There are two firms competing for the orders, we will denote them 1 and 2. None of them has a potential to build both hotels in full. Each firm can offer the building of one hotel or the cooperation on both of them. The investor has to realize the building by force of these firms and he decides on the base of a sealed-bid method. The rules for splitting the orders according to offers are the following:

1. If only one firm bids for a contract on a hotel, it receives it all.

2. If two firms bid for a contract on the same hotel and none of them bids for the second one, the investor offers the cooperation to both firms on both hotels, the profits are split fifty-fifty.

3. If one firm bids for a contract on the whole building of a hotel and the second firm offers a collaboration, than the firm that bids the whole building receives 60% and the second 40% in the case of B, and 80% versus 20% in the case of S. On the building of the other hotel the firms will collaborate fifty-fifty (including splitting the profit).

In every case the total profit of 15 + 9 = 24 milion is split between the firms. Find optimal strategies for both firms.

QUESTION 7.

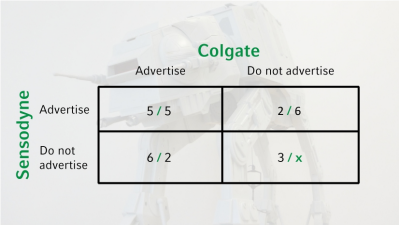
Two firms found themselves in the following price game:

		Firm 2	
		Low	High
Firm 1	Low	40 , 40	65 , 35
	High	35 , 65	60 , 60

CEO of firm 1 wants to get rid of Prisoners' dilemma game and proposed firm 2 to exchange x% of firm's 1 share on the same percentage of firm's 2 shares. What is the minimum of x% (integer) and what payoff should CEO of the firm 1 expect?

QUESTION 8.

Which values can be plugged in for x so that the game turns into a prisoners' dilemma?



QUESTION 9.

What is a strategy according to the working definition of this course?

1. A player's plan of actions in a game.
2. A set of rules a player can choose from.
3. The behaviour of a firm in a certain setting.
4. The best alternatives a player has.

QUESTION 10.

British Airways is the monopolist for domestic flights in the United Kingdom. Lufthansa wants to enter the market. British Airways threatens Lufthansa to start a price war if they enter. Is this a credible threat?

